Lecture 9

Position and momentum on the ring: the Weyl pair

Objectives. *A particle on the ring of Lecture 8 can be read in two ways: by its site, or by its Fourier mode. The two readings play the roles of position and momentum. One unitary, the discrete Fourier transform, exchanges them. No state gives a certain outcome in both readings. More precisely, the numbers of sites and of modes that a state occupies multiply to at least N. Two unitaries label the two readings, and they satisfy Weyl's relation $DT = \zeta TD$. Their products are a basis of all matrices, and for a ring of prime size, readings in $N+1$ bases fix any state. No finite matrices satisfy the canonical relation $[Q,P] = i\hbar\mathbb{1}$ of position and momentum on a line, as a matrix identity. This is where the finite theory stops.*

By the end of this lecture you will be able to:

- Find the site and mode probabilities of a state on the ring, using the discrete Fourier transform.
- Compute with the clock and the shift, using Weyl's relation and the transform that exchanges them.
- Find the supports of a state in both readings, and check the support uncertainty relation.
- Expand a density matrix in clocks and shifts, and say which readings determine its coefficients when N is prime.

9.1 Two readings of a particle on the ring

Take the ring of $N \geq 3$ sites of Lecture 8. Recall its site basis e_x and its Fourier basis

$$f_k = \frac{1}{\sqrt{N}} \sum_x \zeta^{kx} e_x, \qquad \zeta = e^{2\pi i/N}.$$

Each basis is the eigenbasis of an observable, so each gives an ideal measurement.

Definition 9.1 (Site and mode readings)**.** The *site reading* is the measurement with the projections $|e_x\rangle\langle e_x|$, $x \in \mathbb{Z}_N$. In the state $[\psi]$ it gives x with probability

$$p_x = |\psi_x|^2, \quad \text{where} \quad \psi_x = \langle e_x, \psi\rangle .$$

The *mode reading* is the measurement with the projections $|f_k\rangle\langle f_k|$, $k \in \mathbb{Z}_N$. It gives k with probability

$$q_k = |\langle f_k, \psi\rangle|^2 .$$

The site reading tells where the particle is. The mode reading tells how its amplitude winds round the ring. If the sites are a distance a apart, the phase of f_k advances by $2\pi k/N$ over that distance. The phase advance per unit length, $2\pi k/(Na)$, is the *wave number* of f_k. In the next course, *Continuous Systems*, the momentum of a wave is $\hbar$ times its wave number, so k is a label for momentum. It is defined only modulo N, since $f_{k+N} = f_k$.

The amplitudes of the mode reading are the *discrete Fourier transform* of the amplitudes ψ_x:

$$(F\psi)_k = \langle f_k, \psi \rangle = \frac{1}{\sqrt{N}} \sum_{x \in \mathbb{Z}_N} \zeta^{-kx} \psi_x. \tag{9.1}$$

The matrix F has the entries $F_{kx} = N^{-1/2}\zeta^{-kx}$. Its rows are the conjugates of the orthonormal vectors f_k, so F is unitary, and $Ff_k = e_k$. Then $q_k = |(F\psi)_k|^2$. The transform carries the phases ζ^{-kx}, and the basis vectors f_k carry the phases ζ^{+kx}, with the same signs as the Fourier transform of the next course.

The two distributions p and q are not functions of each other. The transform acts on the amplitudes, so their phases enter q. The next example shows two states with one p and two different q.

9.2 The clock and the shift

Each reading has a unitary whose eigenvectors are its basis. For the modes this is the shift T of Lecture 6, $Te_x = e_{x+1}$, with $Tf_k = \zeta^{-k} f_k$ (§ 8.2). For the sites it is the *clock* D:

$$De_x = \zeta^x e_x. \tag{9.2}$$

The eigenvalues ζ^x of D are distinct, and so are the eigenvalues ζ^{-k} of T. So the eigenprojections of D are the projections of the site reading, and those of T the projections of the mode reading.

Theorem 9.2 (The Weyl pair). *The clock and the shift are unitaries on $\mathbb{C}^N$ with*

$$DT = \zeta TD, \qquad D^N = T^N = \mathbb{1}, \tag{9.3}$$

and the transform F exchanges them:

$$FDF^\dagger = T, \qquad FTF^\dagger = D^\dagger. \tag{9.4}$$

Proof. First, $DTe_x = De_{x+1} = \zeta^{x+1}e_{x+1}$ and $TDe_x = \zeta^x e_{x+1}$, so $DT = \zeta TD$. Next, $D^N e_x = \zeta^{Nx} e_x = e_x$ and $T^N e_x = e_{x+N} = e_x$. For the transform, $F^\dagger e_k = f_k$, and

$$Df_k = \frac{1}{\sqrt{N}} \sum_x \zeta^{kx} \zeta^x e_x = f_{k+1},$$

so $FDF^\dagger e_k = FDf_k = Ff_{k+1} = e_{k+1} = Te_k$. Finally, $FTF^\dagger e_k = FTf_k = \zeta^{-k} Ff_k = \zeta^{-k} e_k = D^\dagger e_k$. ∎

So the clock shifts the modes, $Df_k = f_{k+1}$, as the shift shifts the sites, and the transform turns one picture into the other. The relation $DT = \zeta TD$ is *Weyl's relation*. The two unitaries commute up to the phase ζ, and they do not commute. The clock is a function of any observable whose projections are those of the site reading, $D = \sum_x \zeta^x |e_x\rangle\langle e_x|$, and the shift is a function of any observable whose projections are those of the mode reading. If two such observables commuted, so would D and T. So by Theorem 5.2 the site and mode readings are not compatible.

Neither D nor T is an observable. For $N \geq 3$ they are unitary and not self-adjoint, since some of their eigenvalues, such as ζ, are not real. Their eigenvalues only label the readings. We do not take a logarithm to turn them into observables.

9.3 No state is sharp in both

The *support* of a vector $\phi \in \mathbb{C}^N$ is the set of labels at which its components are not zero. Its size is written $|\operatorname{supp}\phi|$. For a state ψ, the site reading can only give the labels in $\operatorname{supp}\psi$, and the mode reading only those in $\operatorname{supp} F\psi$. A state is *sharp* in a reading if one outcome of that reading is certain.

Theorem 9.3 (Support uncertainty). *For every nonzero $\psi \in \mathbb{C}^N$,*

$$|\operatorname{supp}\psi| \cdot |\operatorname{supp} F\psi| \geq N. \tag{9.5}$$

Proof. Divide ψ by its norm. The supports do not change. Let s and t be the sizes of the two supports. Every entry of F has modulus $N^{-1/2}$, so by (9.1), the triangle inequality and the Cauchy–Schwarz inequality over the s sites of $\operatorname{supp}\psi$,

$$|(F\psi)_k| \leq \frac{1}{\sqrt{N}} \sum_{x \in \operatorname{supp}\psi} |\psi_x| \leq \frac{1}{\sqrt{N}} \sqrt{s} \Big(\sum_x |\psi_x|^2 \Big)^{1/2} = \sqrt{\frac{s}{N}} \quad \text{for every } k.$$

Since F is unitary, the q_k add up to 1, and only t of them are nonzero, so

$$1 = \sum_{k \in \operatorname{supp} F\psi} |(F\psi)_k|^2 \leq t \cdot \frac{s}{N}.$$

Hence $st \geq N$. ∎

A state sharp in position, $\psi = e_x$, has $s = 1$, so $t = N$, and every mode is possible. In fact $q_k = 1/N$ for every k. A state sharp in momentum, $\psi = f_k$, is spread over all the sites in the same way. Both states reach the bound. For six sites, states sharp in neither reading reach it too, as the example below shows. Theorem 9.3 is a statement about which outcomes the two readings can give, not about the spread of their values. It is not the uncertainty relation $\Delta Q\, \Delta P \geq \hbar/2$ of the next course, nor the bound on spreads of Theorem 5.7.

Proposition 9.4 (The mode reading is conserved). *Under the Hamiltonian $H = c\mathbb{1} - \kappa(T + T^\dagger)$ of Lecture 8, the mode probabilities q_k of every state are the same at every time. The site probabilities need not be.*

Proof. By Theorem 8.1 each f_k is an eigenvector of H, so by Theorem 4.4 the amplitude $\langle f_k, U_t\psi\rangle$ is $e^{-\mathrm{i}t\varepsilon_k/\hbar}\langle f_k, \psi\rangle$. Its modulus does not change. The walk of § 8.5 shows site probabilities that change in time. ∎

Example 9.5 (Six sites). Take $N = 6$. Figure 9.1 shows the first three states.

1. *One site.* For $\psi = e_0$, $(F\psi)_k = 6^{-1/2}$, so $q_k = \frac{1}{6}$ for every k. Here $s = 1$ and $t = 6$.
2. *Two opposite sites.* For $\psi = \frac{1}{\sqrt{2}}(e_0 + e_3)$,

$$(F\psi)_k = \frac{1}{\sqrt{12}}\big(1 + \zeta^{-3k}\big) = \frac{1}{\sqrt{12}}\big(1 + (-1)^k\big),$$

so $q = (\frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3}, 0)$. Here $s = 2$ and $t = 3$, and $st = 6$. The bound is reached.

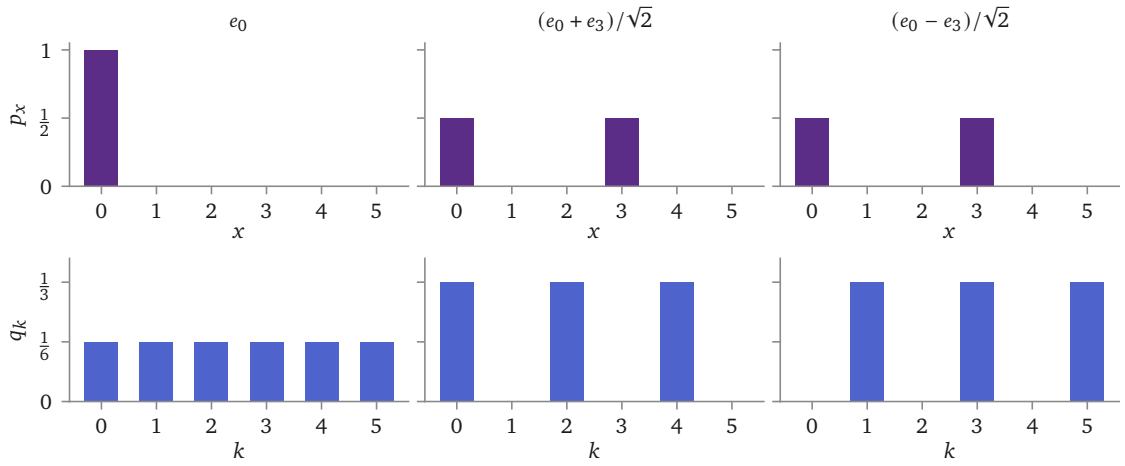


Figure 9.1: The site probabilities p_x (top) and the mode probabilities q_k (bottom) on the ring of six sites, for the three states of Example 9.5. The last two states have the same p and different q.

3. *The same sites, another phase.* For $\psi = \frac{1}{\sqrt{2}}(e_0 - e_3)$, the site probabilities are the same, $p_0 = p_3 = \frac{1}{2}$, but $(F\psi)_k = \frac{1}{\sqrt{12}}\big(1-(-1)^k\big)$, and q puts $\frac{1}{3}$ on each odd k. The site reading does not fix the mode reading.
4. *In time.* Let the state $\frac{1}{\sqrt{2}}(e_0 + e_3)$ evolve under the Hamiltonian of Lecture 8. By Proposition 9.4, q stays $(\frac{1}{3}, 0, \frac{1}{3}, 0, \frac{1}{3}, 0)$. The site probabilities change in time. At $\kappa t/\hbar = \pi/2$ they are $\frac{1}{18}(5, 2, 2, 5, 2, 2)$.

9.4 Every matrix from clocks and shifts

A qubit's density matrix is $\frac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})$. The matrices $\mathbb{1}$, σ_x, σ_y, σ_z are a basis of all 2×2 matrices, and three readings fix the coefficients (Proposition 3.7). On the ring, products of clocks and shifts play the same part. By $DT = \zeta TD$, moving T past D multiplies by ζ^{-1}, so

$$T^s D^r = \zeta^{-rs} D^r T^s, \tag{9.6}$$

and the product of two matrices D^rT^s is a phase times a third. The matrices are compared with the inner product $\mathrm{Tr}(A^\dagger B)$ of *Linear Algebra*, Ex. 7.2.

Theorem 9.6 (A basis of clocks and shifts). *The matrices D^rT^s satisfy*

$$\mathrm{Tr}\big((D^rT^s)^\dagger D^{r'}T^{s'}\big) = N\,\delta_{rr'}\delta_{ss'} \quad \text{for all } r,s,r',s' \in \mathbb{Z}_N.$$

So the N^2 matrices D^rT^s are an orthogonal basis of the $N\times N$ matrices, and every matrix is

$$A = \frac{1}{N}\sum_{r,s\in\mathbb{Z}_N} \mathrm{Tr}\big((D^rT^s)^\dagger A\big)\, D^rT^s.$$

Proof. By (9.6), $(D^rT^s)^\dagger D^{r'}T^{s'} = T^{-s}D^{r'-r}T^{s'}$ is a phase times D^aT^b, with $a = r'-r$ and $b = s'-s$. For $b \neq 0$, T^b moves every site, so a diagonal matrix times T^b has zeros on its diagonal and trace 0. For $b = 0$, $\mathrm{Tr}\,D^a = \sum_x \zeta^{ax}$, and by the geometric sum of § 8.2 this is N for $a = 0$ and 0 otherwise. So the N^2 matrices are orthogonal and nonzero, and the space of $N\times N$

matrices has dimension N^2. Hence they are a basis, and the coefficient of D^rT^s in A is its inner product with A divided by N, its squared norm. ∎

For a density matrix the expansion reads

$$\rho = \frac{1}{N}\sum_{r,s} c_{rs} D^r T^s, \qquad c_{rs} = \mathrm{Tr}\big(\rho (D^r T^s)^\dagger\big),$$

with $c_{00} = 1$. Since ρ is self-adjoint, $c_{-r,-s} = \zeta^{-rs}\bar{c}_{rs}$, and the coefficients carry $N^2 - 1$ real numbers. For $N = 2$, $\zeta = -1$, $D = \sigma_z$, $T = \sigma_x$ and $DT = \mathrm{i}\sigma_y$, and the expansion is $\rho = \frac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})$.

The coefficients can be read. A unitary W with distinct eigenvalues w_j has an orthonormal eigenbasis. A measurement in that basis, repeated on many systems prepared alike, gives the probability p_j of the eigenvector for each w_j, and so $\mathrm{Tr}(\rho W^m) = \sum_j p_j w_j^m$ for every m. The unitary W itself need not be an observable. For $N \geq 3$, the clock, the shift and the unitaries D^uT used below are not observables.

Proposition 9.7 (Tomography on a ring of prime size)**.** *Let N be prime. The site reading and the readings in the eigenbases of the N unitaries D^uT, $u \in \mathbb{Z}_N$, determine every density matrix on $\mathbb{C}^N$.*

Proof. First, $W = D^uT$ has N distinct eigenvalues, for every N. By (9.6), W^N is a phase times $D^{uN}T^N = \mathbb{1}$, say $\gamma\mathbb{1}$, so every eigenvalue is one of the N roots $\lambda_0\zeta^\ell$ of $\lambda^N = \gamma$. Let n_ℓ be their multiplicities. For $0 < m < N$, W^m is a phase times $D^{um}T^m$, whose trace is 0, as in the proof of Theorem 9.6. So $\sum_\ell n_\ell \zeta^{\ell m} = 0$, while $\sum_\ell n_\ell = N$. Multiplying by $\zeta^{-\ell' m}$ and adding over $m = 0, \ldots, N-1$, the geometric sum gives $N n_{\ell'} = N$. So each root occurs once. Hence the reading in the eigenbasis of D^uT gives $\mathrm{Tr}(\rho\,(D^uT)^m)$ for every m, and the site reading, the eigenbasis of D, gives $\mathrm{Tr}(\rho D^r)$. Next, let $(r,s) \neq (0,0)$. If $s = 0$, $D^rT^s = D^r$. If $s \neq 0$, then, since N is prime, s has an inverse s^{-1} in $\mathbb{Z}_N$. With $u = rs^{-1}$, by (9.6), D^rT^s is a phase times $(D^uT)^s$. So every $\mathrm{Tr}(\rho D^rT^s)$, hence every $c_{rs} = \overline{\mathrm{Tr}(\rho D^rT^s)}$, is read, and Theorem 9.6 gives ρ. ∎

The number of readings matches the number of parameters. The $N+1$ readings of N outcomes each give $(N+1)(N-1) = N^2 - 1$ independent probabilities, the number of real parameters of ρ. For $N = 2$ they are the readings along z, x and y of Proposition 3.7. For $N = 3$ they are four bases. For prime N these $N+1$ bases are *mutually unbiased*, which means that a state of one basis gives each outcome of another with probability $1/N$ (Exercise Sheet 9 for $N = 3$). Primality is used only to divide by s. For $N = 4$ this family of bases does not suffice, and other constructions are needed.

9.5 Where the finite theory stops

For a particle on a line, position and momentum are observables Q and P that satisfy, on a suitable set of states that the next course specifies,

$$QP - PQ = \mathrm{i}\hbar\mathbb{1}.$$

The clock and the shift look like exponentials of such a pair. On the line, the unitaries $e^{-\mathrm{i}aP/\hbar}$ translate by a, as T translates by one site, and $e^{\mathrm{i}bQ/\hbar}$ multiply by a phase that grows with the position, as D does. They satisfy

$$e^{-\mathrm{i}aP/\hbar}e^{\mathrm{i}bQ/\hbar} = e^{-\mathrm{i}ab/\hbar}e^{\mathrm{i}bQ/\hbar}e^{-\mathrm{i}aP/\hbar},$$

the form of Weyl's relation that the next course proves. The finite relation $DT = \zeta TD$ holds for matrices on $\mathbb{C}^N$. The relation for the generators Q and P does not, as the next proposition shows.

Proposition 9.8 (No finite position and momentum)**.** *There are no $n \times n$ matrices Q and P, for any $n \geq 1$, with $QP - PQ = \mathrm{i}\hbar\mathbb{1}$.*

Proof. By the cyclicity of the trace, $\mathrm{Tr}(QP - PQ) = \mathrm{Tr}(QP) - \mathrm{Tr}(PQ) = 0$, while $\mathrm{Tr}(\mathrm{i}\hbar\mathbb{1}) = \mathrm{i}\hbar n \neq 0$. ∎

This is *Linear Algebra*, Prop. 14.24. The canonical relation needs operators on a space of infinite dimension, and that is where the next course begins. It starts with a particle on a line, and with the uncertainty relation $\Delta Q\, \Delta P \geq \hbar/2$, which has no finite form here.

The physics

Weyl wrote down the clock and the shift in 1927, as a finite model of position and momentum, and Schwinger proved Theorem 9.6 in 1960. Wootters and Fields showed in 1989 that readings in mutually unbiased bases determine a state with the fewest kinds of reading, and that such complete sets of $N + 1$ bases exist when N is a power of a prime. For $N = 6$ no complete set is known. The same pair appears in signal processing, where a signal sampled at N equally spaced times is transformed by F into N discrete frequency components. There Theorem 9.3 says that a short pulse has a broad spectrum. Donoho and Stark proved it in this form in 1989, and used it to recover signals with missing samples.

Position and momentum on the ring

Readings: the site reading, with probabilities $p_x = |\psi_x|^2$, and the mode reading, with $q_k = |(F\psi)_k|^2$. The transform F is unitary (Definition 9.1). *Labels:* the clock D and the shift T, unitaries with $DT = \zeta TD$, exchanged by F (Theorem 9.2). *Uncertainty:* $|\mathrm{supp}\,\psi| \cdot |\mathrm{supp}\,F\psi| \geq N$ (Theorem 9.3). *Basis:* the N^2 matrices $D^r T^s$, orthogonal, so $\rho = \frac{1}{N}\sum_{r,s} c_{rs} D^r T^s$ (Theorem 9.6). For prime N, $N{+}1$ readings determine ρ (Proposition 9.7). *Limit:* no matrices satisfy $QP - PQ = \mathrm{i}\hbar\mathbb{1}$ (Proposition 9.8).

Summary.

- *The Weyl pair*: the clock and the shift label the site and mode readings, satisfy $DT = \zeta TD$, and are exchanged by the discrete Fourier transform (Theorem 9.2).
- *Support uncertainty*: no state is sharp in both readings (Theorem 9.3).
- *A basis of clocks and shifts*: the N^2 matrices $D^r T^s$ are an orthogonal basis, and for prime N the site reading and the eigenbases of the $D^u T$ determine every state (Theorem 9.6, Proposition 9.7).
- *No finite position and momentum*: $QP - PQ = \mathrm{i}\hbar\mathbb{1}$ has no matrix solution (Proposition 9.8).

Parts I and II have treated one system at a time: a qubit, then n levels, with their symmetries, their multiplets and the ring. Lecture 10 begins Part III with two systems at once.

References

Banks, *Quantum Mechanics: An Introduction*, §3.2.1, pp. 64–68, Ex. 3.14, p. 87, and Ex. 6.6, p. 161. *Linear Algebra for Physics*, Prop. 14.24. Weyl, *Zeitschrift für Physik* 46, 1 (1927). Schwinger, *Proceedings of the National Academy of Sciences* 46, 570 (1960). Wootters and Fields, *Annals of Physics* 191, 363 (1989). Donoho and Stark, *SIAM Journal on Applied Mathematics* 49, 906 (1989).